

3(a)	any three from: stimulates an <u>immune response</u> ; (usually) protein / glycoprotein / polysaccharides ; A polypeptides I glycolipids recognised by / activates / stimulates, and named immune system cell ; e.g. phagocyte / macrophage / neutrophil (B- / T-) lymphocyte I plasma cell A <i>idea of</i> leads to production of antigen-presenting cell <i>allow correct example of response by named cell</i> cause production of (specific) antibodies / bind to (specific) antibodies ; A plasma cells releasing (specific) antibody A complementary shape to (antigen binding site on) antibody A complementary to receptors on lymphocytes example of location of non-self antigen ; I whole cells as non-self antigens e.g. surface of / AW, pathogens / infected cells / tumour cells / non-body cells / foreign cells bacterial, cell wall / flagella viral, capsid / protein coat / envelope AVP ; e.g. high molecular weight can be a toxin produced by a pathogen	3
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Question	Answer	Marks																								
3(b)	one mark each correct column = 4 marks if 0 or 1 mark gained, check for correct row to max 2 <i>In the whole table:</i> • if only ticks are used, assume blanks are crosses • if only crosses are used, assume blanks are ticks <i>If there are ticks and crosses and blanks in the whole table, R the columns or rows with the blank(s)</i> <table><tr><th rowspan="2">description of cell</th><th colspan="4">cell of the immune system</th></tr><tr><th>B-lymphocyte</th><th>plasma cell</th><th>T-helper cell</th><th>T-killer cell</th></tr><tr><td>able to go through a number of cell cycles (clonal expansion)</td><td>✓</td><td>x</td><td>✓</td><td>✓</td></tr><tr><td>main role is to secrete cytokine during an immune response</td><td>x</td><td>x</td><td>✓</td><td>x</td></tr><tr><td>present during a primary immune response to a virus</td><td>✓ ;</td><td>✓ ;</td><td>✓ ;</td><td>✓ ;</td></tr></table>	description of cell	cell of the immune system				B-lymphocyte	plasma cell	T-helper cell	T-killer cell	able to go through a number of cell cycles (clonal expansion)	✓	x	✓	✓	main role is to secrete cytokine during an immune response	x	x	✓	x	present during a primary immune response to a virus	✓ ;	✓ ;	✓ ;	✓ ;	4
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3(c)	A = cytokinesis ; A division of the cytoplasm / cytoplasmic division B = interphase ; C = DNA replication ; A semi-conservative replication / production of sister chromatid A DNA, synthesis / AW I chromatid duplication / chromatid replication	3																								